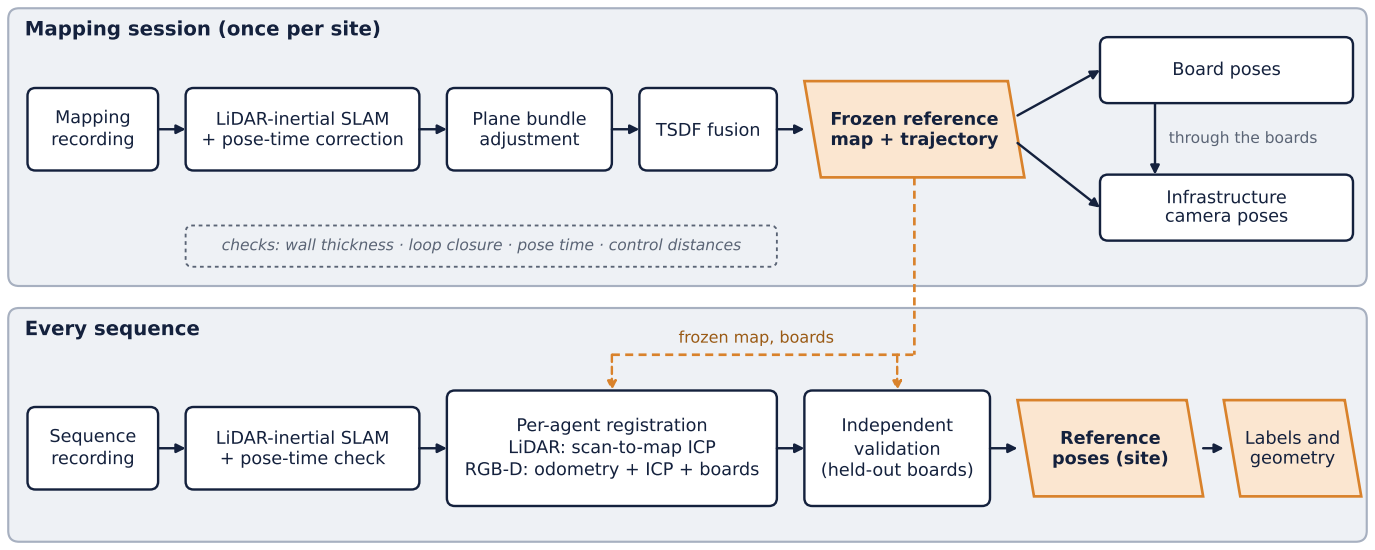
Mapping session (once per site)
Mapping recording
LiDAR-inertial SLAM + pose-time correction
Plane bundle adjustment
TSDF fusion
Frozen reference map + trajectory
Board poses
through the boards
Infrastructure camera poses
checks: wall thickness · loop closure · pose time · control distances
Every sequence
frozen map, boards
Sequence recording
LiDAR-inertial SLAM + pose-time check
Per-agent registration
LiDAR: scan-to-map ICP
RGB-D: odometry + ICP + boards
Independent validation (held-out boards)
Reference poses (site)
Labels and geometry